

Label Correction Enhanced Dual-Stream Multiple Instance Learning for Weakly-Supervised Depression Detection in Speech

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Introduction

Backgrounds

- Social prejudice and limited medical resources on depression diagnosis call for effective speech-based Automatic Depression Detection (**ADD**) systems, considering the availability for spoken signals to convey affective information.
- However, existing speech-based ADD works fail to jointly address **inaccurate and inexact labels** for ADD in weakly-supervised cases.

Sun Y, Zhou Y, Xu X, et al. Weakly-supervised depression detection in speech through self-learning based label correction[J]. IEEE Transactions on Audio, Speech and Language Processing, 2025, 33: 748-758.

Introduction

Motivation

- First, speech-based ADD tends to trust the judgments from annotators or subjective questionnaires, ignoring **inaccurately-supervised** cases for incorrect labels.
- Second, depressive traits in low-quality audio samples may be confined to subtle segments, possibly resulting in **inexactly-supervised** cases in training.

*In this regard, we propose a weakly-supervised ADD approach in speech using **Label Correction enhanced Dual-stream Multiple Instance Learning (LC-DMIL)**.*

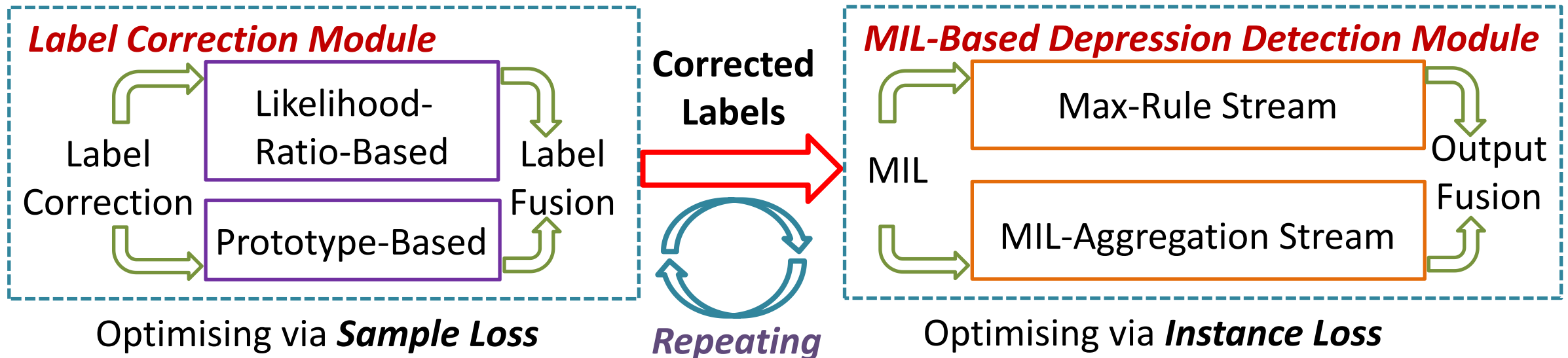
- For the **first issue**, the proposed approach involves **a label correction module** to rectify possibly inaccurate training labels.
- For the **second issue**, the approach includes **a dual-stream Multi-Instance Learning (MIL) strategy**, in order to effectively capture instance-level subtle depressive traits.

Methodology

The Network

The proposed LC-DMIL consists of two modules:

- First, the **label correction module** aims to perform sample-level label correction through a fusion of correction strategies.
- Then, the **MIL-based depression detection module** performs instance-level learning within samples using the corrected labels, through setting two parallel streams for MIL (**max-rule** and **MIL-aggregator streams**).



Methodology

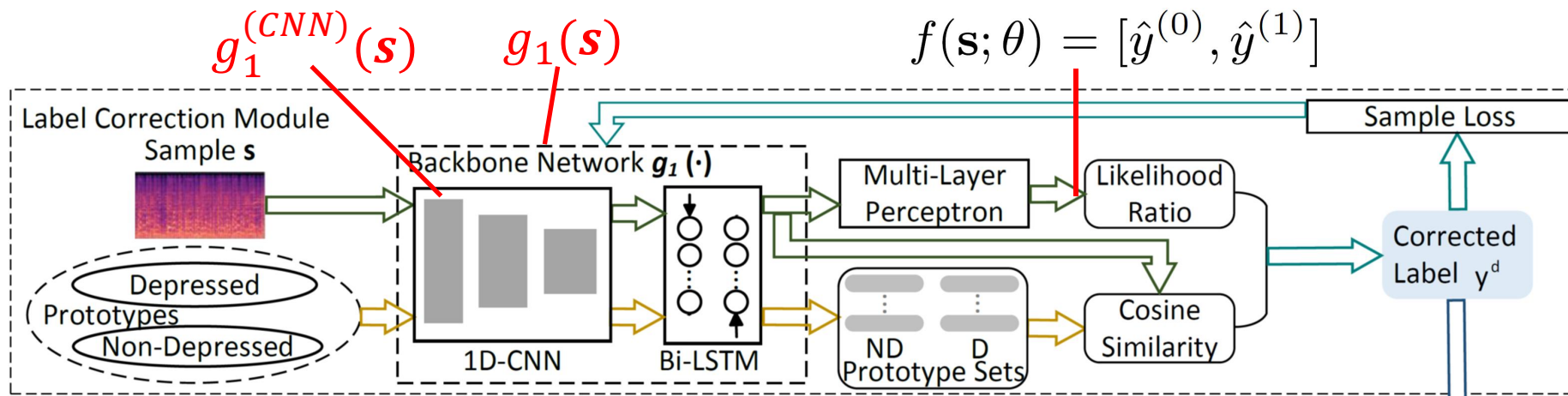
The Label Correction Module

- For the label correction module, we perform label correction through fusing corrected labels from two label-correction strategies: The **likelihood-ratio-based** and **prototype-based** strategies. The backbone network $g_1(\cdot)$ consists of a 1D-CNN layer and a Bi-LSTM layer.

$$\tilde{y}^d = \text{softmax}([1 - (1 - \gamma)y^R - \gamma y^P, (1 - \gamma)y^R + \gamma y^P]) = [1 - y^d, y^d]$$

likelihood-ratio-based

prototype-based



Methodology

Label Correction: *Likelihood-Ratio-Based Strategy*

- For the **likelihood-ratio-based** strategy, we compare sample s 's likelihood ratio with a threshold.
- If the likelihood ratio is less than or equal to the threshold, we flip the sample's origin label to obtain likelihood-ratio-based corrected label y^R , otherwise the label remains unchanged.

Threshold: $\delta(n^{(Eq)})$ — # Training Epochs

$$R(f, s, y) = (\hat{y}^{(0)})^{1-2y} (\hat{y}^{(1)})^{2y-1}$$

Output (Depression)

Output (Non-Depression)

→ Label flipping if: $R(f, s, y) \leq \delta(n^{(Eq)})$

Methodology

Label Correction: *Prototype-Based Strategy*

- For the prototype-based strategy, we define a similarity matrix on all the N training samples, and then, define each training sample's density.
- We set the samples with the top q highest density values for each class to construct prototype sets.
- In this way, we calculate the similarity between the sample $\mathbf{s}$ and class c 's prototype set, and then, predict the corrected label.

Similarity Matrix: $\mathbf{S} \in \mathbb{R}^{N \times N}$, with elements $S_{k,j}$

$$\rightarrow \textcircled{1} \text{Den}_k = \sum_{j=1}^N \text{sgn}(S_{k,j} - \mathbf{S}^{(M)}) \quad \rightarrow \textcircled{2} \text{Sim}^{(c)}(\mathbf{s}) = \frac{1}{q} \sum_{l=1}^q \cos(g_1(\mathbf{s}), g_1(\mathbf{s}_l^{(c)}))$$

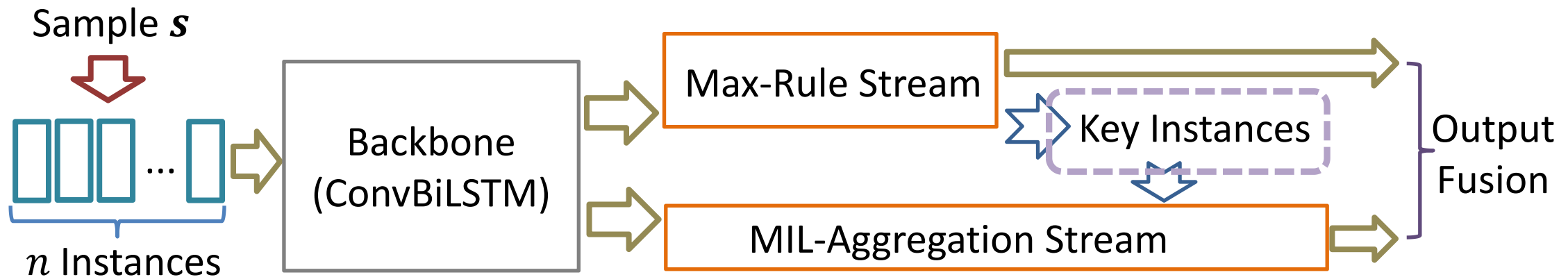
Sign Function Median of matrix $\mathbf{S}$'s elements

$$\rightarrow \textcircled{3} y^P = \arg \max_{c \in \{0,1\}} \text{Sim}^{(c)}(\mathbf{s})$$

Methodology

The MIL-Based Depression Detection Module

- In the MIL-based depression detection module, we employ a ConvBiLSTM backbone ($g_2(\cdot)$) for instance-level depression detection, using the same structure as in the label correction module.
- For a sample s with its corrected label y^d , we split the sample into n instances as a bag ($\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$), resulting in the backbone's L -dimensional output embeddings for $\mathbf{x}_i$ ($\mathbf{a}_i = g_2(\mathbf{x}_i)$).
- Then, the embeddings are fed to two streams to perform MIL, through jointly considering two MIL strategies.



Methodology

MIL-Based Depression Detection: *Max-Rule Stream*

The max-rule stream aims to obtain **key instances**, via utilising an MLP ($f^{(\text{IMLP})}(\cdot)$) as the instance-level classifier, for two output dimensions of depression and non-depression.

- Then, the MLP is followed by a max-rule function on the two dimensions, with the outputs for the labels 0 and 1 via finding maximum elements.
- Thus, we obtain the max-rule stream's output $p^{(M)}$, and set corresponding key instances for the two classes (as $\tilde{\mathbf{a}}^{(0)}$ and $\tilde{\mathbf{a}}^{(1)}$), respectively.

$$\textcircled{1} f^{(\text{IMLP})}(\mathbf{s}) = [f^{(\text{IMLP})}(\mathbf{a}_1), f^{(\text{IMLP})}(\mathbf{a}_2), \dots, f^{(\text{IMLP})}(\mathbf{a}_n)] \equiv [f_0^{(\text{IMLP})}(\mathbf{s})^\top, f_1^{(\text{IMLP})}(\mathbf{s})^\top]^\top$$

$$\rightarrow \textcircled{2} p_0^{(M)}(\mathbf{s}) = \max(f_0^{(\text{IMLP})}(\mathbf{s})) \text{ and } p_1^{(M)}(\mathbf{s}) = \max(f_1^{(\text{IMLP})}(\mathbf{s})),$$

$$\rightarrow \textcircled{3} \mathbf{p}^{(M)} = [p_0^{(M)}(\mathbf{s}), p_1^{(M)}(\mathbf{s})]$$

Methodology

MIL-Based Depression Detection: *MIL-Aggregator Stream*

- The MIL-aggregator stream aggregates the instances' embeddings using the key instances. We transform an arbitrary instance's embedding $\mathbf{a}_j$ into a query vector $\mathbf{q}_j$ and a value vector $\mathbf{v}_j$.
- Then, we calculate the similarity between the j th instance and the class- c key instances using $\tilde{\mathbf{a}}^{(c)}$ ($c = 0, 1$) from the max-rule stream. Further, we obtain linear combination for the value vectors using the similarities as the weights, and map $\mathbf{E}$ to the output of the MIL-aggregator stream ($p^{(A)}(\mathbf{s})$).

$$\textcircled{1} D(\mathbf{q}_j, \tilde{\mathbf{q}}^{(c)}) = e^{\tilde{\mathbf{q}}^{(c)\top} \mathbf{q}_j} \left(\sum_{k=1}^n e^{\tilde{\mathbf{q}}^{(c)\top} \mathbf{q}_k} \right)^{-1}$$

$$\rightarrow \textcircled{2} \mathbf{E} = \left[\sum_{k_0=1}^n D(\mathbf{q}_{k_0}, \tilde{\mathbf{q}}^{(0)}) \mathbf{v}_{k_0}, \sum_{k_1=1}^n D(\mathbf{q}_{k_1}, \tilde{\mathbf{q}}^{(1)}) \mathbf{v}_{k_1} \right]$$

$$\rightarrow \textcircled{3} p^{(A)}(\mathbf{s}) = \mathbf{w}_A \mathbf{E} + \mathbf{b}_A$$

$$\left\{ \begin{array}{l} \mathbf{q}_j = \tanh(\mathbf{W}_q \mathbf{a}_j + \mathbf{b}_q) \\ \mathbf{v}_j = \text{ReLU}(\mathbf{W}_v \mathbf{a}_j + \mathbf{b}_v) \\ \tilde{\mathbf{q}}^{(c)} = \tanh(\mathbf{W}_q \tilde{\mathbf{a}}^{(c)} + \mathbf{b}_q) \end{array} \right\}$$

Methodology

The Loss

- For the **label correction module**, a sample loss is calculated as $L^{(S)}$ using the samples' labels and prediction in an m -sample training batch, through combining the classification loss $L^{(C)}$ and entropy loss $L^{(E)}$.

$$L^{(S)} = \frac{1}{m} \sum_{i=1}^m \left(L^{(C)}(f(\mathbf{s}_i; \theta), y_i^d) + \alpha L^{(E)}(f(\mathbf{s}_i; \theta)) \right) \quad (\alpha=0.1)$$

- Then, for the **MIL-based depression detection module**, we first combine the results from the two streams to calculate the output $p(\mathbf{s})$, and employ an instance loss on a batch of m_0 instances.

$$p(\mathbf{s}) = \text{softmax}(\mu p^{(M)}(\mathbf{s}) + (1 - \mu)p^{(A)}(\mathbf{s}))$$

$$L^{(I)} = \frac{1}{m_0} \sum_{i_0=1}^{m_0} \left(L_0^{(C)}(p(\mathbf{s}_{i_0}), \boxed{z_{i_0}^d}) + \alpha L_0^{(E)}(p(\mathbf{s}_{i_0})) \right) \quad \text{Instances' labels (same as their corresponding samples)}$$

Experiments

Database

- We employ clinical interviews with 142 subjects from the DAIC-WOZ database, utilising the training set including 107 utterances (31 depression), and 35 utterances (12 depression) for testing.
- The subjects' responses are individually extracted, and split into non-overlapping segments of 6s for each sample.
- Then, to simulate the weak supervision, we swap labels of samples with **PHQ-8 scores between 7 and 12** in the training set, as these samples with marginal scores are highly likely to be incorrectly labelled.
- Through splitting and label swapping, we accordingly obtain 7.7k samples for training (2.3k depression) and 2.9k samples for testing (1.2k depression).

Experiments

Results - Comparison with Related Approaches

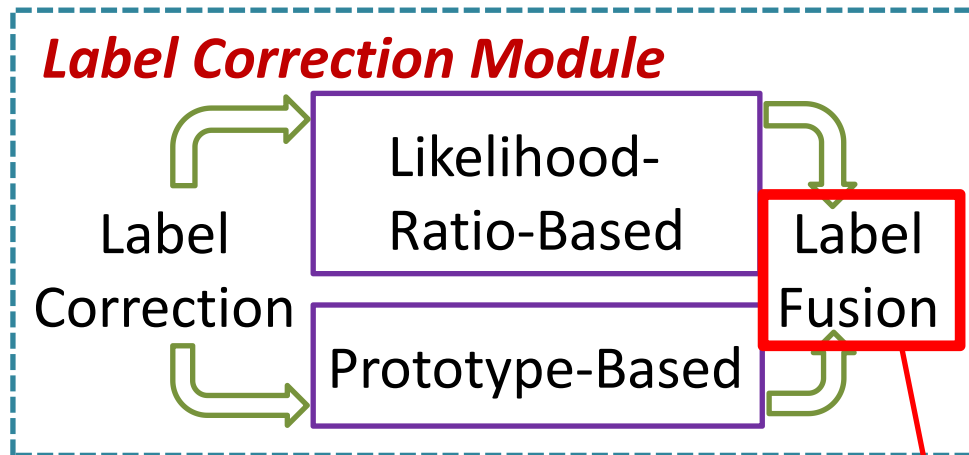
- First, we show the results for the proposed LC-DMIL with **related approaches**, which indicates the better performance for the proposed approach, especially compared with our previously proposed SLLC.

Approaches \ Metrics	UAR	F1-Score
DepAudioNet [3]	0.574	0.580
ConvBiLSTM [28]	0.580	0.567
SpeechFormer [29]	0.576	0.549
ComParE [30]	0.550	0.540
CMT-SMER (Audio) [31]	0.540	0.563
SLLC [6]	0.614	0.613
LC-DMIL	0.651	0.642

Experiments

Results

- Then, we present the results when setting the **label-correction fusion weight** to different values, for the previous SLLC without MIL and the proposed LC-DMIL. The results indicate the effectiveness for the inclusion of MIL.



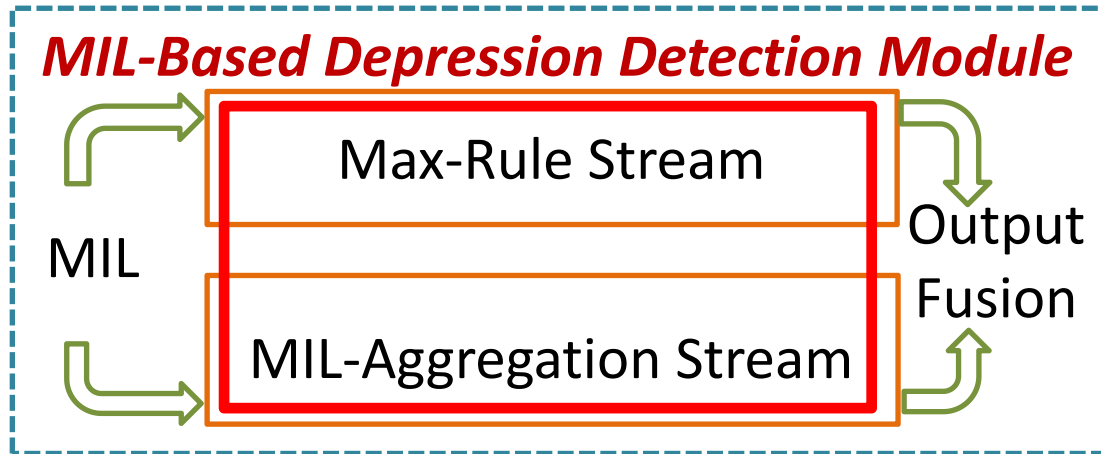
Setups \ Metrics	UAR	F1-Score
SLLC (R) ($\gamma = 0$)	0.603	0.595
SLLC (P) ($\gamma = 1$)	0.591	0.588
SLLC ($\gamma = 0.7$)	0.560	0.552
SLLC ($\gamma = 0.3$)	0.614	0.613
LC-DMIL ($\gamma = 0.7$)	0.620 (+0.060)	0.617 (+0.065)
LC-DMIL ($\gamma = 0.3$)	0.651 (+0.037)	0.642 (+0.029)

Fusion Weight γ

Experiments

Results

- Afterwards, we focus on investigating **different dual-stream setups** in the MIL-based depression detection module, which shows the effectiveness for the dual-stream setup in the MIL for the proposed approach.

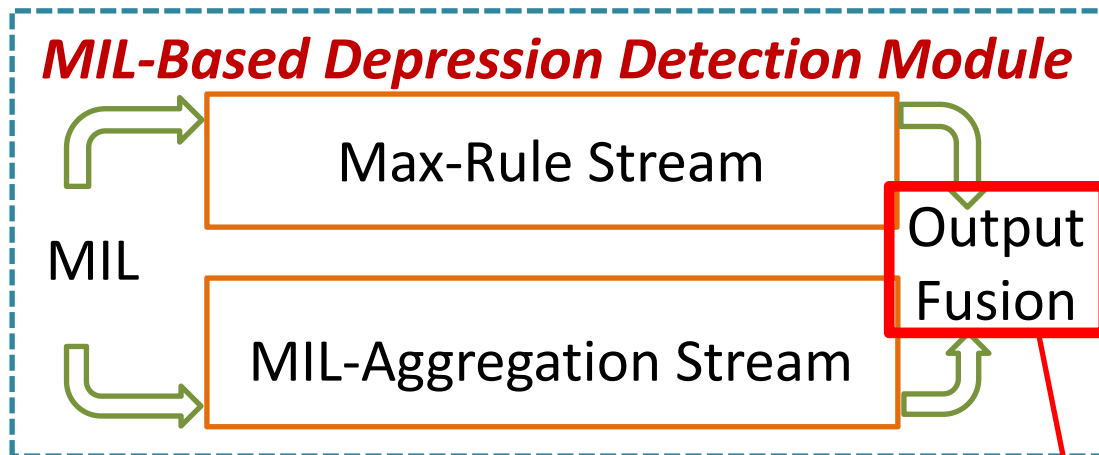


MIL Strategies \ Metrics	UAR	F1-Score
Mean-Rule [25]	0.626	0.622
Weighted-Rule [25]	0.631	0.633
Transformer-Based [35]	0.637	0.631
Attention-Based [36]	0.633	0.625
Max-Rule (as in LC-DMIL)	0.632	0.614
MIL-Aggregator (as in LC-DMIL)	0.636	0.633
Dual-Stream (LC-DMIL)	0.651	0.642

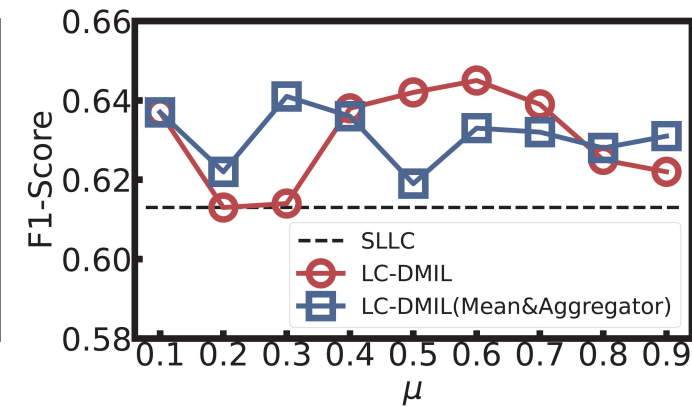
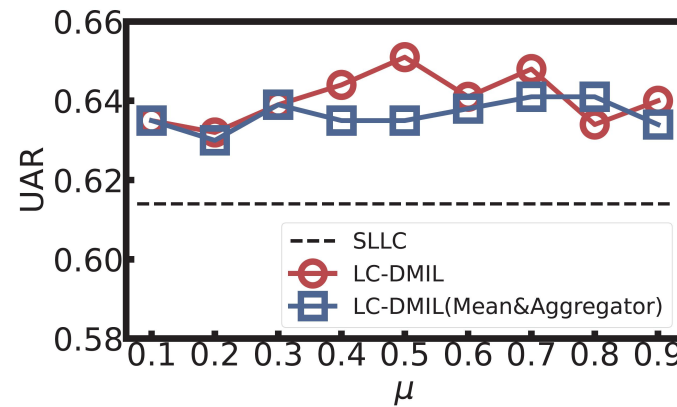
Experiments

Results

- We further investigate the **MIL's dual-stream output fusion** through setting the weight μ to different values in the MIL-based depression detection module, with different stream settings of SLLC, the LC-DMIL using mean-rule strategy instead of the max-rule one, and the proposed LC-DMIL.
- The comparisons show the effectiveness for fusing the max-rule and MIL-aggregator streams in the MIL.



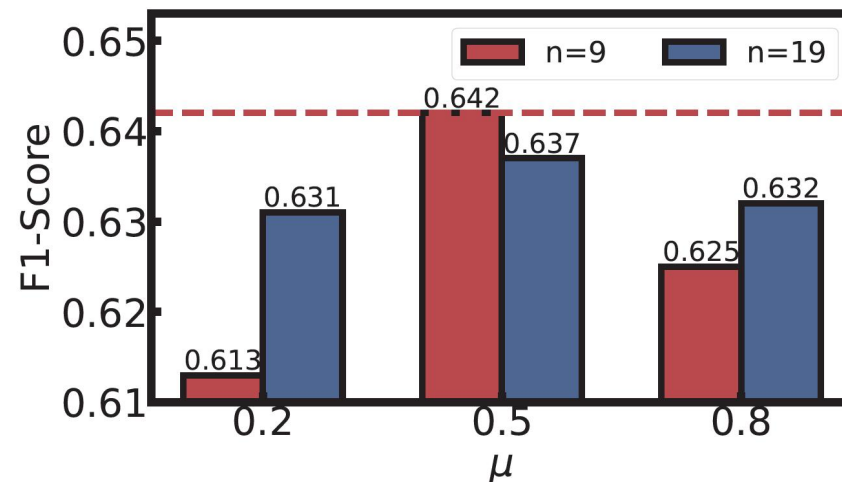
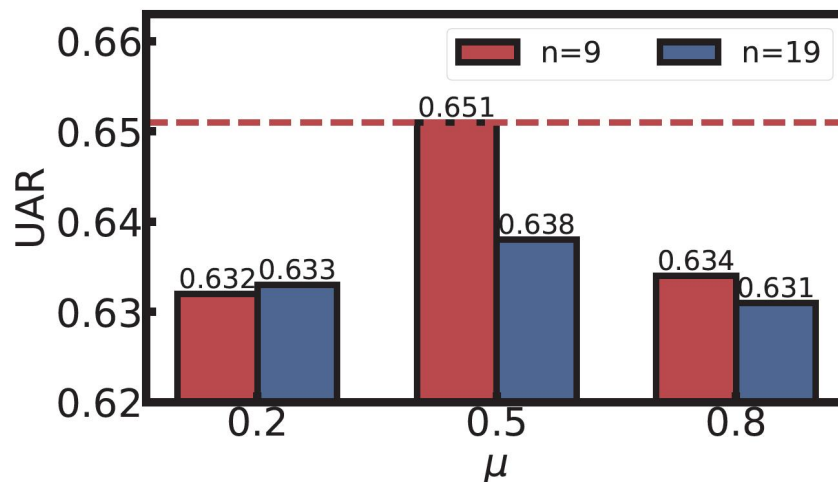
Fusion Weight μ



Experiments

Results

- To explore the influence from **the number of instances in the MIL**, we make comparison for the proposed approach, when setting the instance number n to 9 and 19.
- The results imply that, more instances may not lead to better results, possibly due to the scale on representing effective segments for depression detection.



Experiments

Results

- Finally, we show the **ablation results for label correction and MIL** based on the proposed LC-DMIL on two datasets (noted as 'AVEC 2014 / DAIC-WOZ'), in which we employ random label swapping for the AVEC 2014 dataset.
- The results indicate the effectiveness of the proposed modules.

Setups \ Metrics	UAR	F1-Score
w/o Both (ConvBiLSTM)	0.644 / 0.580	0.643 / 0.567
MIL-Only (DMIL)	0.628 / 0.569	0.629 / 0.562
LC-Only (SLLC)	0.657 / 0.614	0.655 / 0.613
w/ LC&MIL (LC-DMIL)	0.670 / 0.651	0.668 / 0.642

Conclusion & Future Works

Conclusion

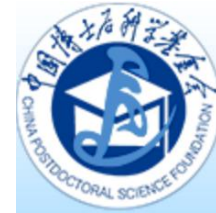
- *We proposed the Label Correction enhanced Dual-stream Multiple Instance Learning (LC-DMIL) approach for weakly-supervised depression detection in speech.*
- *The proposed approach comprises a label correction module focusing on **inaccurate supervision** via fused label correction strategies, and an MIL-based depression detection module employing dual-stream MIL to address **inexact supervision**.*

Future Works

- *Then, future works may focus on including generated data for weakly-supervised speech-based depression detection, and weakly-supervised learning for other affective computing issues.*

Thank you!

Q & A



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